Night-Sky Photo Workflow: Stacking, Star Removal & Meteor Hunting

A practical guide to processing astrophotos with free tools and an AI assistant (Claude Code) doing the heavy lifting. Works on Mac and Windows.

What this guide covers:

1. **Stacking deep-sky photos.** Combine many short shots into one clean, colorful image, including data from **Unistellar** smart telescopes.
2. **Removing the stars.** This makes the nebula or galaxy stand out; you can add the stars back at lower strength.
3. **Finding meteors.** Go through a whole night of camera frames automatically and keep only the frames with a meteor, not a plane or satellite.
4. **Timelapses and star trails** that play smoothly on phones and laptops.

You don’t need to be a programmer. The AI can write and run the scripts. Your job is to say what you want and tell it when the results look wrong.

Quick glossary

Term	What it means
Sub / sub-frame	One single exposure. A finished image is made from many subs.
Stacking	Lining up and averaging many subs. Noise cancels out and faint detail builds up.
Registration	The “lining up” step before stacking: shifting and rotating each sub so the stars overlap.
Stretching	Stacked images look almost black. Stretching brightens the faint stuff without blowing out the stars. Think of it as an extreme curves adjustment.
Starless image	The same photo with the stars digitally removed, so only the nebula or galaxy is left.
Plate solving	The software works out exactly where in the sky the photo points by matching its star pattern.
Color calibration (SPCC)	Uses star catalog data to set “true” colors, like a scientifically accurate white balance.
Bayer pattern	The layout of red, green and blue pixels on a color sensor (RGGB, GBRG, etc.). If the software uses the wrong one, the colors come out wrong.

Term	What it means
FITS	The standard file format for astronomy images. Telescope cameras and smart telescopes save in this format.
Raw files (NEF, CR3, ARW, DNG)	Camera raw formats. DNG is Adobe's universal raw format.
Field rotation	On mounts that move up/down and left/right (like most smart telescopes), the sky slowly rotates in the frame during a session.
Bitrate / CRF	How much data per second a video uses. Higher means bigger files that are harder for devices to play.

What you'll need (Mac & Windows)

Everything here is free.

App	What it's for	Mac	Windows
Claude Code	AI assistant that writes and runs the scripts	Install the **Claude desktop app** from claude.ai/download	Same
Siril	Stacking, color calibration, stretching	Download from siril.org	Download from siril.org
StarNet2	Star removal	Download the Mac version from starnetastro.com and unzip it into your home folder	Download the Windows version from starnetastro.com and unzip it to `C:\StarNet2`
Python 3.12	Runs the meteor-finding scripts	See the Homebrew note below, then run `brew install python@3.12`	Download from python.org and **tick "Add Python to PATH"** during install
ExifTool	Reads the capture time from each photo	`brew install exiftool`	Download from exiftool.org
FFmpeg	Turns frames into videos	`brew install ffmpeg`	Download from ffmpeg.org (the "gyan.dev" builds are easiest)
Adobe DNG Converter *(optional)*	Converts raws for apps that can't read them	Adobe download	Same

Mac users: what's `brew`? Homebrew is a free tool for installing apps from the Terminal. Install it once from brew.sh by copying its one-line command into Terminal. After that, every `brew install …` line above just works.

Windows users: shortcut. If you're comfortable with the command line, `winget install Python.Python.3.12`, `winget install OliverBetz.ExifTool` and `winget install Gyan.FFmpeg` install three of

these in one go.

Easy mode: you can also just ask Claude, *"Install Python 3.12, ExifTool and FFmpeg for me and set up a folder for a meteor-finding script."* It will run the commands and install the Python add-ons (numpy, OpenCV, etc.) itself.

Connect StarNet2 to Siril: in Siril, open **Preferences → Miscellaneous**, find **StarNet executable**, and point it at the `starnet2` file (`starnet2.exe` on Windows) in the folder you unzipped.

Storage: a night of shooting on a high-res camera can be **50–100 GB**, and processing makes temporary copies. Have at least twice that free before you start.

Part 1: Stacking deep-sky photos (Siril + StarNet2)

This works for any camera or telescope: DSLR/mirrorless on a tracker, a dedicated astro camera, or a smart telescope.

The basic steps

1. **Load your subs** into Siril and convert them to a sequence.
2. **Register** them so the stars line up.
3. **Stack** with outlier rejection. This automatically removes satellite trails, plane trails and hot pixels that show up in only a few subs.
4. **Crop** the ragged edges left over from alignment.
5. **Plate solve**, then run **color calibration (SPCC)** for accurate color.
6. **Remove the background gradient.** This takes out the uneven glow from light pollution or the moon.
7. **Denoise gently.** Half strength is a good start; full strength looks smudgy.
8. **Stretch** to bring out the faint detail. Use the *linked* option (see tips below).
9. **Add saturation** to taste.
10. **Remove the stars with StarNet2.** You get two layers: starless, and stars only.
11. **Blend the stars back in** at around 30–50%. The target becomes the hero and the stars stay subtle.

Save: a PNG or JPEG for sharing, a **16-bit TIFF** for editing in Photoshop or Lightroom, and the stacked `.fit` file as your untouched master.

Tips and common problems

- **Always use a "linked" stretch.** An unlinked stretch brightens each color channel separately, which throws the color balance off.
- **Hazy or smoky starless image?** It's usually too much denoise plus saturation. Denoise less, and run background removal again *on the starless image*.
- **Want more control than auto-stretch?** Try Siril's **GHS** (Generalized Hyperbolic Stretch) tool.

- **Download the SPCC data** from Siril's settings the first time. Without it, color calibration falls back to a generic sensor profile.
- **Soft or blurry result?** Check how high your target was. Objects low on the horizon look softer because you're shooting through more atmosphere.
- **Quick tracking check:** play your subs back in order like a flipbook. If the image wanders, your tracking or polar alignment was off.
- **Harmless warnings:** Siril sometimes shows a "no suitable data" message on denoise, and Mac users may see Python crash warnings. Both are safe to ignore.

Part 1b: Stacking Unistellar smart-telescope data

Unistellar telescopes (eVscope, Odyssey, etc.) stack automatically in the app. If you stack the **individual raw frames** yourself in Siril, you can usually get noticeably better color, less noise and more detail.

The raw files

Unistellar raw frames are **FITS** files (`.fits` / `.fit`), one per sub. A single target is usually hundreds to thousands of small files. Put each target's frames in its own folder before you start.

Unistellar-specific steps in Siril

1. **Check the color (Bayer) pattern before anything else.** The pattern in Unistellar file headers can be wrong. Odyssey frames, for example, are labeled **RGGB** but are really **GBRG**. If your first test stack looks strangely tinted (green/magenta or swapped colors), go to Siril's debayer settings, tick the option to **ignore the header pattern**, and try **GBRG**.
2. **Debayer and register** with rotation allowed. Unistellar uses an up/down–left/right mount, so the field rotates during the session. Siril's normal star-alignment handles this.
3. **Stack with rejection.** Unistellar sessions often have hundreds of subs, so rejection does a great job of removing satellite trails.
4. **Crop generously.** Field rotation leaves wedge-shaped edges that need cutting off.
5. **Continue with steps 5–11 above:** plate solve, color calibration, background removal, stretch, star removal.

Unistellar tips

- **Throw out bad frames.** Frames hit by clouds, wind gusts or a bump will be blurry or faint. Siril can sort frames by quality after registration, so drop the worst ones.
- **Don't trust the pointing info in the headers too much.** If plate solving fails, solve using the target's name or coordinates instead of the position stored in the file.
- **Long sessions are worth it.** Unistellar subs are short, so a few thousand of them gives a much cleaner result than a few hundred.
- **The telescope repoints now and then during long sessions.** The framing may shift in jumps rather than drifting smoothly. Registration fixes this, but you'll lose a bit more at the edges.

Part 2: Finding meteors in a night of photos

If you leave a camera shooting all night during a meteor shower, you can end up with thousands of frames. A script can find the few with a meteor and throw out planes, satellites, clouds and trees. Ask Claude to build one using the approach below.

How a meteor finder works

1. **Sort the frames by capture time** using ExifTool, and remove duplicate or empty files (memory cards sometimes have them).
2. **Use the preview JPEG built into each raw file.** It's much faster than decoding the full raw, and plenty for detection.
3. **Compare each frame with the ones next to it** to find new streaks of light.
4. **Ignore the foreground** (trees, buildings, horizon) using a mask.
5. **Check how thin each streak is.** Meteors are razor thin (1–2 pixels); cloud edges are much fatter. This is often the single most useful test.
6. **Weed out planes and satellites:**
    - **Planes** have blinking lights, so their trail looks dashed and often has red or green in it.
    - **Satellites** show up in several frames in a row along the same straight line.
    - **Meteors** are a single smooth streak in a single frame, often getting brighter or changing color along the way.
7. **Skip twilight frames** at dusk and dawn.
8. **Make contact sheets** (grids of thumbnails) and a spreadsheet of candidates, so you can check them by eye quickly.
9. **Make circled copies** of the meteor frames, so the streak is easy to spot when you share them.

Tips and common problems

- **Some newer raw formats** (like Nikon's "High Efficiency" NEF) can't be opened by much free software. The built-in preview JPEG works fine for detection.
- **Converting to DNG isn't always a win.** For compressed raw formats, DNGs can come out *larger* and lose camera-specific settings. Keep your originals.
- **No automatic filter is perfect.** Strict plane filters will also throw out some real meteors. Also ask for an **"all streaks" folder** and give it a quick look yourself.
- **Damaged files happen.** Make sure the script skips unreadable frames instead of crashing.
- **Test it.** Ask the script to add fake meteors to some frames and check how many it finds. That tells you whether your settings are too strict or too loose.
- **Green sky?** That's probably **airglow**, a natural glow in the upper atmosphere. It isn't your lens, and it's not aurora unless the space-weather **Kp index** is high.

Part 3: Timelapses & star trails

From the same night of frames you can make:

- a **timelapse** of the whole night (4K at 24 fps looks great)
- **star-trail videos** where trails build up over time, or a "comet-tail" version where they fade behind each star
- a single **star-trail photo** combining the whole night
- a **"stars-only" trail** where planes and satellites fade out instead of cluttering the image
- a **meteor composite**: all the night's meteors placed into one photo

Making videos that actually play smoothly

High-res night-sky timelapses often **stutter or freeze**. There are two usual causes:

1. **The file is too heavy.** Sensor grain makes video files huge, often hundreds of Mbps, which phones and laptops can't keep up with. A light noise reduction and slightly more compression bring it down to tens of Mbps with no visible difference.
2. **The wrong video format for the resolution.** At 4K-and-above resolutions, the common H.264 format often won't play on Macs, iPhones or iPads. Use **HEVC (H.265)** instead.

Here's a good starting FFmpeg command. You can also just ask Claude to "make a 4K HEVC timelapse from this folder":

```
ffmpeg -framerate 24 -i frames/%05d.jpg \
  -vf "hqdn3d=3:3:0:0" \
  -c:v libx265 -crf 20 -preset slow -tag:v hvc1 -pix_fmt yuv420p \
  timelapse_4k.mp4
```

- `hqdn3d=3:3:0:0` is a light noise reduction that works **within** each frame only. Avoid noise reduction that blends frames together, because it erases meteors (they only appear in one frame).
- `-crf 20` sets the quality. Lower numbers mean higher quality and bigger files; 18–23 is a good range.
- `-tag:v hvc1` makes the video play on Apple devices.
- **Windows:** to play HEVC in the built-in Photos/Movies apps, install **HEVC Video Extensions** from the Microsoft Store, or just use **VLC** (free).

Tips:

- **Make one version with your edit** (contrast, gamma) and **one natural version**, then pick the better one.
- **A star-trail video doubles as a tracking check.** On a tracked shot, the stars shouldn't trail.
- **Save disk space** by converting raws in batches instead of all at once.

Tips for working with an AI assistant on this

- **Write down what you want first.** A short notes file (“find meteors, ignore planes, give me contact sheets”) keeps things on track across sessions.
- **Ask for things you can look at:** contact sheets, before/after pairs, a spreadsheet of results. That’s much faster than opening files one by one.
- **Push back when something looks off.** “You’re missing meteors,” “this looks hazy” and “the video stutters” are exactly the feedback that gets you to a good result.
- **Ask it to explain.** If a step or setting doesn’t make sense, ask “why?” It’s a good way to learn the processing as you go.
- **Want to look at results on your phone?** Ask for a simple web page with before/after toggles.